

DOKUMENTACJA GAIA

1. Opis i Wymagania

system, w którym inteligentne kontenery z czujnikami zapełnienia komunikują się autonomicznie z siecią śmieciarek, przekazując im informacje o potrzebie opróżnienia. Śmieciarki muszą się między sobą koordynować, aby ustalić, która z nich jest najbliższą i może wykonać zadanie w danym momencie. Dzięki temu problem nadmiernych przejazdów, opóźnień w opróżnianiu kontenerów oraz niepotrzebnych operacji zostaje zminimalizowany, co skutkuje bardziej efektywnym systemem gospodarki odpadami oraz wyższą satysfakcją klientów.

- System umożliwia powiadomienie śmieciarki o konieczności opróżnienia konkretnego kontenera
- System umożliwia wybór najbliższej niepełnej śmieciarki do danego zadania
- System umożliwia monitorowanie zapełnienia kontenerów
- System umożliwia monitorowanie stanu śmieciarek (zajęty, pełny, wolny)
- System umożliwia terminowy wywóz śmieci (kontener nie może być zapełniony dłużej niż 1 dzień)
- Śmieciarki bardziej zapełnione są priorytetizowane -> unikanie sytuacji gdy nagle wszystkie śmieciarki są pełne i żadna nie może nic wywieźć

2. Model Ról

Role: Container

Description	Entity representing a smart waste container equipped with fill level sensors that communicates its status to the system.
Responsibilities	Liveness: (ContainerToDispatchSystemNotification)^ω Safety: UniqueID(ContainerID) != Null FillLevel >= 0 FillLevel <= 100 Timestamp != Null
Permissions	READ: ContainerID, FillLevel, Timestamp WRITE: ContainerID, FillLevel, Timestamp CONSUME: ContainerID, FillLevel, Timestamp PRODUCE: ContainerID, FillLevel, Timestamp
Activities	
Protocols	ContainerToDispatchSystemNotification

Role: GarbageTruck

Description	Autonomous vehicle responsible for collecting waste from containers, capable of receiving assignment and reporting status.
Responsibilities	Liveness: (DispatchSystemToGarbageTruckAssignment . DispatchSystemToGarbageTruckStatusUpdate . DispatchSystemToGarbageTruckPriorityAssignment)^ω Safety: UniqueID(GarbageTruckID) != Null Status in ['Available', 'Busy', 'Maintenance'] Timestamp != Null FillLevel >= 0 FillLevel <= 100 PriorityLevel >= 0
Permissions	READ: GarbageTruckID, Status, Timestamp, FillLevel, PriorityLevel WRITE: GarbageTruckID, Status, Timestamp, FillLevel, PriorityLevel CONSUME: GarbageTruckID, Status, Timestamp, FillLevel, PriorityLevel PRODUCE: GarbageTruckID, Status, Timestamp, FillLevel, PriorityLevel
Activities	
Protocols	DispatchSystemToGarbageTruckAssignment, DispatchSystemToGarbageTruckStatusUpdate, DispatchSystemToGarbageTruckPriorityAssignment

Role: DispatchSystem

Description	Centralized system that coordinates garbage truck assignments, monitors container fill levels, and manages task prioritization.
Responsibilities	Liveness: (AssignGarbageTruck MonitorContainerStatus MonitorGarbageTruckStatus EnforceDeadline PrioritizeGarbageTrucks)^ω Safety: UniqueID(ContainerID) != Null FillLevel >= 0 FillLevel <= 100 Timestamp != Null UniqueID(GarbageTruckID) != Null Status in ['Available', 'Busy', 'Maintenance'] DeadlineTimestamp != Null
Permissions	READ: ContainerID, FillLevel, Timestamp, GarbageTruckID, Status, DeadlineTimestamp WRITE: ContainerID, FillLevel, Timestamp, GarbageTruckID, Status, DeadlineTimestamp CONSUME: ContainerID, FillLevel, Timestamp, GarbageTruckID, Status, DeadlineTimestamp PRODUCE: ContainerID, FillLevel, Timestamp, GarbageTruckID, Status, DeadlineTimestamp
Activities	AssignGarbageTruck, MonitorContainerStatus, MonitorGarbageTruckStatus, EnforceDeadline, PrioritizeGarbageTrucks
Protocols	DispatchSystemToContainerStatusUpdate, DispatchSystemToGarbageTruckStatusUpdate, DispatchSystemToContainerDeadlineEnforcement, DispatchSystemToGarbageTruckPriorityAssignment

3. Model Interakcji

Global	Name	Initiator	Responder	Inputs	Outputs	Description
Garbage container requirement	ContainerToDispatchSystemNotification	Container	DispatchSystem	ContainerID, FillLevel, Timestamp		A container sends a notification to the dispatch system when it requires emptying.
Dispatch system assigns the closest available garbage truck to a container	DispatchSystemToGarbageTruckAssignment	DispatchSystem	GarbageTruck	ContainerID, GarbageTruckID, EstimatedArrivalTime		Dispatch system assigns the closest available garbage truck to a container requiring emptying.
Container fill status update	DispatchSystemToContainerStatusUpdate	DispatchSystem	Container	ContainerID	FillLevel, Timestamp	Dispatch system retrieves current fill level status from containers for monitoring purposes.
Garbage truck status update	DispatchSystemToGarbageTruckStatusUpdate	DispatchSystem	GarbageTruck	GarbageTruckID	Status, Timestamp	Dispatch system retrieves current status of garbage trucks for coordination purposes.
Deadline-based enforcement	DispatchSystemToContainerDeadlineEnforcement	DispatchSystem	Container	ContainerID, DeadlineTimestamp		Dispatch system enforces deadline for container emptying to prevent overflow.
Garbage truck assignment based on fill	DispatchSystemToGarbageTruckPriorityAssignment	DispatchSystem	GarbageTruck	GarbageTruckID, FillLevel	PriorityLevel	Dispatch system assigns priority to garbage trucks based on their fill levels.

4. Model Usage

Service	Agent	In/Out	Pre/Post Condition	Derived From
NotifyDispatchSystemOfContainerStatus	ContainerAgent	IN: ContainerID, FillLevel, Timestamp OUT:	PRE: FillLevel >= 90 AND ContainerID != Null AND Timestamp != Null POST: ContainerID != Null AND FillLevel >= 0 AND FillLevel <= 100 AND Timestamp != Null	ContainerToDispatchSystemNotification
AssignGarbageTruckToContainer	DispatchSystemAgent	IN: ContainerID, GarbageTruckID, EstimatedArrivalTime OUT:	PRE: ContainerID != Null AND GarbageTruckID != Null AND EstimatedArrivalTime != Null AND Status in ['Available', 'Busy', 'Maintenance'] POST: Status in ['Available', 'Busy', 'Maintenance']	DispatchSystemToGarbageTruckAssignment
RetrieveContainerStatus	ContainerAgent	IN: ContainerID OUT: FillLevel, Timestamp	PRE: ContainerID != Null POST: FillLevel >= 0 AND FillLevel <= 100 AND Timestamp != Null	DispatchSystemToContainerStatusUpdate
RetrieveGarbageTruckStatus	GarbageTruckAgent	IN: GarbageTruckID OUT: Status, Timestamp	PRE: GarbageTruckID != Null POST: Status in ['Available', 'Busy', 'Maintenance'] AND Timestamp != Null	DispatchSystemToGarbageTruckStatusUpdate
EnforceContainerDeadline	DispatchSystemAgent	IN: ContainerID, DeadlineTimestamp OUT:	PRE: ContainerID != Null AND DeadlineTimestamp != Null AND Timestamp < DeadlineTimestamp AND FillLevel >= 90 POST: ContainerID != Null AND Timestamp != Null	DispatchSystemToContainerDeadlineEnforcement
AssignGarbageTruckPriority	DispatchSystemAgent	IN: GarbageTruckID, FillLevel OUT: PriorityLevel	PRE: GarbageTruckID != Null AND FillLevel >= 0 AND FillLevel <= 100 POST: PriorityLevel >= 0	DispatchSystemToGarbageTruckPriorityAssignment
MonitorContainerFillLevels	DispatchSystemAgent	IN: ContainerID OUT: ContainerID, FillLevel, Timestamp	PRE: ContainerID != Null AND FillLevel >= 0 AND FillLevel <= 100 POST: All containers monitored and updated with latest FillLevel and Timestamp	MonitorContainerStatus
MonitorGarbageTruckStatuses	GarbageTruckAgent	IN: GarbageTruckID OUT:	PRE: GarbageTruckID != Null AND Status in ['Available', 'Busy', 'Maintenance'] POST: All garbage trucks monitored and updated with latest Status and Timestamp	MonitorGarbageTruckStatus
EnforceCollectionDeadline	DispatchSystemAgent	IN: ContainerID, DeadlineTimestamp OUT:	PRE: ContainerID != Null AND Timestamp != Null AND Timestamp < DeadlineTimestamp AND FillLevel >= 90 POST: Deadline enforced for all containers with FillLevel > 90	EnforceDeadline
PrioritizeGarbageTrucksByFillLevel	DispatchSystemAgent	IN: GarbageTruckID, FillLevel OUT:	PRE: GarbageTruckID != Null AND FillLevel >= 0 AND FillLevel <= 100 POST: Garbage trucks prioritized based on current FillLevel	PrioritizeGarbageTrucks

5. Model Znajomości

GarbageTruckAgent



ContainerAgent



DispatchSystemAgent

